Stroma World — Gene State Representation & Combination Rules v1.0

> **Design principle:** The simplest system that preserves PDAC biology. No floats. No multipliers. Just states and logic gates.

Part 1: Gene State Alphabet

Every gene in the simulation exists in exactly one of four states:

State	Symbol	Meaning	When to assign
WT	0	Wild-type / normal function	Default state; gene behaving as in healthy pancreatic tissue
GOF	+	Gain-of-function	Constitutive activation, amplification, or overexpression beyond normal regulation
LOF	−	Loss-of-function	Deletion, inactivating mutation, epigenetic silencing, or functional suppression
UNK	?	Unknown / not yet determined	Placeholder during initialization or when patient data is incomplete

Rules

1. **Every gene gets exactly one state.** No partial states, no gradients, no expression levels.
2. **UNK is not a simulation state.** It exists only during setup. Before a run begins, every ? must resolve to 0, +, or −. Resolution rule: ? → WT unless the EA or scenario explicitly sets it otherwise.

3. **State is fixed for the lifetime of a cell.** Genes do not change state mid-simulation. Phenotype changes come from *signal-dependent* Boolean logic acting on fixed gene states, not from gene state transitions. (Cycle 2+ may add stochastic state transitions for modeling acquired resistance.)
4. **GOF and LOF are defined relative to normal pancreatic ductal epithelium** (for tumor genes) or **normal pancreatic stellate cells** (for stromal genes). Not relative to some abstract baseline.

Part 2: Default PDAC Genotype

The "canonical late-stage PDAC" genotype that most simulation runs should start with:

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TUMOR CELL DEFAULT GENOTYPE
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KRAS    +   (GOF — constitutive, ~95% of PDAC)
MYC     +   (GOF — transcriptionally upregulated)
EGFR    0   (WT  — present but not the primary driver)
TP53    –   (LOF — mutant/deleted, ~75%)
BCL-XL  +   (GOF — overexpressed)
BAX     0   (WT  — present but functionally outcompeted by BCL-XL)
CDKN2A  –   (LOF — homozygous deletion, ~90%)
SMAD4   –   (LOF — deleted/mutant, ~55%)
RB1     0   (WT  — intact but functionally inactivated by CDKN2A loss)
ZEB1    0   (WT  — inducible by signals, not constitutive)
CDH1    0   (WT  — expressed until ZEB1 activates)
MMP2    0   (WT  — inducible)
HIF1A   0   (WT  — activated by hypoxia, not constitutive)
NRF2    0   (WT  — activated by stress, not constitutive)
ABCB1   0   (WT  — activated by NRF2, not constitutive)
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STROMAL CELL (PSC/CAF) DEFAULT GENOTYPE
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ACTA2   0   (WT  — quiescent; activates on TGF-β/SHH signal)
GLI1    0   (WT  — activates on SHH signal)
HAS2    0   (WT  — activates when CAF is activated)
COL1A1  0   (WT  — activates when CAF is activated)
TGFB1   0   (WT  — secretion rate is signal-dependent)
SHH     –   (LOF — stromal cells do not secrete SHH; only tumor does)
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Why so many WT?

The distinction between **gene state** and **gene activity** is critical:

- **Gene state** = structural capacity (is the gene intact or broken?)
- **Gene activity** = current functional output (is the gene doing something right now?)

ZEB1 is WT (the gene is intact and functional) but *inactive* at baseline. It becomes active when TGF-β or HIF1A signals arrive. This is handled by the Boolean network, not by gene state. Gene state answers "can this gene work?" Boolean logic answers "is this gene working right now?"

Part 3: Gene State → Signal Contribution

Before combining effects, each gene in its current state emits a **signal contribution** to each behavioral axis. This is not a number — it's a qualitative push.

Push	Symbol	Meaning
STRONG UP	⬆⬆	Dominant driver of this axis
UP	⬆	Contributing push upward
NEUTRAL	⊖	No contribution to this axis
DOWN	⬇	Contributing push downward
STRONG DOWN	⬇⬇	Dominant suppression of this axis
CONDITIONAL	⬆? or ⬇?	Push exists only when a specific signal is present (see condition column)

Full Signal Contribution Table

Only non-neutral contributions are listed. If a gene/axis pair is absent, it's ⊖.

GROWTH DRIVE

Gene	State	Push	Condition
KRAS	+	⬆⬆	— (constitutive)
MYC	+	⬆⬆	—
EGFR	+	⬆	—
CDKN2A	–	⬆	— (brake removal reads as growth push)
SMAD4	–	⬆?	Only when TGF-β present (lost brake reads as growth under TGF-β)
ZEB1	active	⬇	When ZEB1 is ON (go-vs-grow tradeoff)
HIF1A	active	⬇?	Acute hypoxia slows cycling; chronic adaptation may restore it

DEATH RESISTANCE

Gene	State	Push	Condition
TP53	–	⬆⬆	—
BCL-XL	+	⬆⬆	—
BAX	0	⬇	Pro-death push (but outcompeted by BCL-XL when BCL-XL = +)
KRAS	+	⬆	Via PI3K/AKT
ZEB1	active	⬆	Anoikis resistance
HIF1A	active	⬆	Autophagy, BCL-XL upregulation
NRF2	active	⬆	ROS detox
ABCB1	active	⬆⬆?	Only when drug is present

CELL CYCLE BRAKING (higher = more responsive to arrest signals)

Gene	State	Push	Condition
CDKN2A	–	⬇⬇	—
TP53	–	⬇⬇	—
SMAD4	–	⬇	TGF-β cannot trigger arrest
RB1	func. inact	⬇	Convergence node disabled
MYC	+	⬇⬇	Overrides remaining checkpoints
KRAS	+	⬇	Partial checkpoint override

MOTILITY / INVASION

Gene	State	Push	Condition
ZEB1	active	⬆⬆	—
CDH1	repressed	⬆⬆	When ZEB1=ON
MMP2	active	⬆⬆	ECM degradation
SMAD4	–	⬆?	TGF-β drives invasion via SMAD-independent arm
TP53	– (GoF)	⬆	Gain-of-function integrin recycling
HIF1A	active	⬆	Hypoxia-driven migration
ACTA2	active [S]	⬆	CAF contractility remodels ECM tracks

PRO-STROMA SIGNALING

Gene	State	Push	Condition
TGFB1	secreted	⬆⬆	— (main barrier builder)
SHH	+ [T]	⬆⬆	— (paracrine to stroma)
KRAS	+	⬆⬆	Drives TGFB1 + SHH secretion
GLI1	active [S]	⬆	CAF proliferation + ECM
ACTA2	active [S]	⬆⬆	Full CAF activation
HAS2	active [S]	⬆⬆	HA production
COL1A1	active [S]	⬆⬆	Collagen production
HIF1A	active	⬆	CTGF, secondary stroma signals
MMP2	active	⬇	ECM degradation opposes buildup

IMMUNE-EVASION TENDENCY

Gene	State	Push	Condition
TGFB1	secreted	⬆⬆	Treg induction, CD8 suppression
KRAS	+	⬆	GM-CSF, myeloid recruitment
ZEB1	active	⬆	PD-L1 upregulation
HIF1A	active	⬆	Adenosine pathway
COL1A1	active [S]	⬆	Physical T cell exclusion
HAS2	active [S]	⬆	Physical barrier to infiltration

Part 4: Combination Rules

The Core Problem

Multiple genes push the same axis. How do you combine ⬆⬆ + ⬆ + ⬇ into a single behavioral outcome without numbers?

The Rule: Dominance Voting with Veto

Step 1 — Collect all active contributions for the axis. "Active" means the gene is in the relevant state AND any signal condition is met. Ignore conditional pushes whose condition is not satisfied.

Step 2 — Apply the Dominance Rule.

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AXIS OUTCOME = strongest active signal, modulated by opposition.
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Concretely:

Active Pushes	Outcome	Name
All neutral	**BASELINE**	Axis at normal tissue levels
Any ⬆ or ⬆⬆, no ⬇	**HIGH** or **VERY HIGH**	Uncontested upward drive
Any ⬇ or ⬇⬇, no ⬆	**LOW** or **VERY LOW**	Uncontested suppression
Mixed ⬆⬆ and ⬇	**HIGH**	Strong up wins over weak down
Mixed ⬆ and ⬇	**MODERATE**	Contested; partial effect
Mixed ⬆⬆ and ⬇⬇	**MODERATE**	Titans cancel; residual context decides
Single ⬆⬆ alone	**HIGH**	One dominant driver
Multiple ⬆⬆	**VERY HIGH**	Converging dominant drivers (ceiling)

Step 3 — Apply Veto Rules. Certain gene states create hard logical constraints that override voting:

Veto Rule	Effect	Rationale
CDKN2A=– AND TP53=– AND MYC=+	Cell Cycle Braking → **NONE**	All three major brakes destroyed + override active. Arrest is impossible regardless of other signals.
BCL-XL=+ AND BAX=0	Death Resistance floor → **HIGH**	Anti-apoptotic dominant. Apoptosis requires external intervention to overcome this.
SMAD4=– AND TGFB1 present	TGF-β brake on tumor → **DISABLED**	TGF-β cannot arrest this cell. Route to stroma activation remains open.
ZEB1=active AND CDH1=repressed	Motility floor → **HIGH**	Full EMT. Cell is mesenchymal regardless of other adhesion signals.
ABCB1=active AND drug absent	ABCB1 death resistance contribution → **ZERO**	Efflux pump has nothing to pump.

Step 4 — Map outcome to PhysiCell phenotype category.

Outcome	PhysiCell Mapping
VERY HIGH	Parameter at maximum biologically plausible value
HIGH	Parameter at upper quartile of literature range
MODERATE	Parameter at midpoint of literature range
BASELINE	Parameter at normal tissue value
LOW	Parameter at lower quartile of literature range
VERY LOW	Parameter at minimum / near-zero
NONE	Parameter hard-set to zero (or functionally equivalent)

> This is the **only** step where numbers enter. The qualitative logic above selects a *category*. The actual parameter value within that category comes from the literature-backed parameter table (Appendix A of the spec). The combination rules never produce a number — they produce a bin, and the bin has a pre-set value.

Part 5: Worked Examples

Example 1: Growth Drive of a Canonical PDAC Tumor Cell

Genotype: KRAS=+, MYC=+, EGFR=0, CDKN2A=–, SMAD4=–

Collect contributions to Growth Drive:

- KRAS=+ → ⬆⬆
- MYC=+ → ⬆⬆
- EGFR=0 → ➖ (WT, not overexpressed)
- CDKN2A=– → ⬆ (brake removal)
- SMAD4=– → ⬆? (conditional on TGF-β; assume present) → ⬆

Active pushes: ⬆⬆, ⬆⬆, ⬆, ⬆. No opposition. → **Multiple ⬆⬆ with supporting ⬆ → VERY HIGH.**

PhysiCell: `cycle.transition_rate` set to maximum plausible PDAC proliferation rate.

Example 2: Death Resistance Under Drug Treatment

Genotype: TP53=–, BCL-XL=+, BAX=0, NRF2=active, ABCB1=active **Condition:** Drug is present.

Collect contributions to Death Resistance:

- TP53=– → ⬆⬆
- BCL-XL=+ → ⬆⬆
- BAX=0 → ⬇ (pro-death push, but...)
- NRF2=active → ⬆
- ABCB1=active + drug present → ⬆⬆

Active pushes: ⬆⬆, ⬆⬆, ⬆⬆, ⬆ vs. ⬇. Veto check: BCL-XL=+ AND BAX=0 → floor is HIGH regardless. → **Three ⬆⬆ with supporting ⬆, weak opposition (⬇) → VERY HIGH.**

PhysiCell: death.apoptosis_rate at minimum. Drug sensitivity modifier at lowest tier. This cell is nearly unkillable. The EA must find a way around this.

Example 3: Motility of a Tumor Cell in Hypoxic Stroma

Genotype: ZEB1=inactive (baseline), SMAD4=–, TP53=– **Signals:** TGF-β present, O_2 below threshold → HIF1A=active

Boolean network step (happens before combination):

- HIF1A=ON → activates ZEB1
- ZEB1=ON → represses CDH1

Now collect contributions to Motility:

- ZEB1=active → ⬆⬆
- CDH1=repressed → ⬆⬆
- SMAD4=– + TGF-β present → ⬆ (SMAD-independent invasion arm)
- TP53=– → ⬆ (GoF integrin recycling)
- HIF1A=active → ⬆

Active pushes: ⬆⬆, ⬆⬆, ⬆, ⬆, ⬆. No opposition. Veto check: ZEB1=active AND CDH1=repressed → floor is HIGH. → **VERY HIGH.**

PhysiCell: motility.speed at maximum, cell_cell_adhesion at minimum. This cell is fully mesenchymal and actively invading.

Note the causal chain: Stroma → hypoxia → HIF1A → ZEB1 → EMT → invasion. The stroma the tumor built is now driving the tumor to become more aggressive. This emergent feedback is exactly what the simulation needs to capture.

Example 4: Pro-Stroma Signaling — Does the Barrier Self-Assemble?

Tumor genotype: KRAS=+, SMAD4=– **Stromal cell state:** ACTA2=inactive (quiescent PSC) **Signals at start:** None.

Turn 1 — Tumor secretes:

- KRAS=+ drives TGFB1 and SHH secretion (this is a genotype→secretion rule, not a combination)
- TGF-β and SHH diffuse into stroma

Turn 2 — Stromal cell receives:

- TGF-β arrives → ACTA2 activates (PSC → CAF transition)
- SHH arrives → GLI1 activates

Turn 3 — Now combine PSS for the activated CAF:

- ACTA2=active → ⬆⬆
- GLI1=active → ⬆
- HAS2 activates (downstream of ACTA2) → ⬆⬆
- COL1A1 activates (downstream of ACTA2) → ⬆⬆
- CAF begins secreting its own TGF-β → positive feedback

Active pushes: ⬆⬆, ⬆⬆, ⬆⬆, ⬆. No opposition. → **VERY HIGH pro-stroma output.**

ECM density rises → drug diffusion drops → hypoxia increases → HIF1A activates in tumor → tumor becomes more invasive and drug-resistant → tumor secretes more stress signals → stroma responds with more ECM.

This is the barrier self-assembling. The combination rules produce it from gene states and signal logic without any explicit "build stroma" instruction. If your simulation doesn't produce this cascade, the wiring is wrong.

Example 5: SHH Inhibition — Reproducing the Vismodegib Failure

Intervention: EA sets SHH=LOF in tumor cells (simulating vismodegib blocking SHH signaling).

Effect on stroma:

- SHH signal eliminated → GLI1 no longer activates in stroma
- But TGF-β is still present (KRAS still drives it)
- ACTA2 still activates via TGF-β alone (SHH is not required, just additive)
- Stroma is *partially* reduced (GLI1 contribution removed) but not eliminated

Effect on PSS combination for CAF: Before: ACTA2=⬆⬆, GLI1=⬆, HAS2=⬆⬆, COL1A1=⬆⬆ → VERY HIGH After: ACTA2=⬆⬆, GLI1=—, HAS2=⬆⬆, COL1A1=⬆⬆ → still HIGH (one ⬆ removed but three ⬆⬆ remain)

But the stroma also restrains the tumor. Reduced stroma → improved drug diffusion BUT ALSO reduced physical containment. Tumor proliferation is VERY HIGH and no longer compressed. If the intervention didn't include a cytotoxic drug alongside SHH inhibition:

- Stroma partially opens → tumor expands into freed space → worse outcome.

This matches the vismodegib trial result. The combination rules + the self-assembly cascade correctly predict that removing one stromal signal without killing the newly-uncontained tumor makes things worse.

Part 6: Implementation Checklist

- □ Define gene state enum: {WT, GOF, LOF, UNK}
- □ Assign default PDAC genotype for tumor cells
- □ Assign default quiescent genotype for stromal cells
- □ Implement Boolean network (signal → gene activity updates)
- □ After Boolean update, collect active contributions per axis
- □ Apply dominance voting to get qualitative outcome
- □ Apply veto rules to enforce hard constraints
- □ Map outcome category to PhysiCell parameter bin
- □ Look up actual parameter value from literature table
- □ Verify self-assembly test (Example 4) produces stroma barrier
- □ Verify vismodegib test (Example 5) produces tumor acceleration

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Part 7: What This System Cannot Do (and Doesn't Need To)

Limitation	Why it's acceptable for v1.0
No continuous expression levels	Axes capture the qualitative phenotype. Quantitative tuning is the EA's job via the parameter bins.
No mid-run gene state changes	Acquired resistance and clonal evolution are Cycle 2+ problems. Fixed genotypes are sufficient for testing stroma-targeting interventions.

Limitation	Why it's acceptable for v1.0
No epistasis beyond veto rules	Full epistatic modeling requires data we don't have. Dominance voting + vetoes capture the strongest interactions. We can add more vetoes as validation reveals missing ones.
No stochastic state assignment	Every cell of a given type starts with the same genotype. Intra-tumor heterogeneity is a Cycle 3 problem.
Immune-evasion axis is tracked but not simulated	We compute IET outcomes for future use but no immune cells exist in Cycle 1. This is bookkeeping, not wasted effort.

Four states. Six axes. Dominance voting. Five veto rules. No floating-point combination logic anywhere. The numbers enter only at the final mapping step, and they come from the literature table, not from the combination rules.